

# Bayesian Attribution Assessment (Infrastructure & Malware Reuse)

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Prepared by: Automated analytic workflow (ChatGPT) • Classification: Unclassified / For analytic use

## 1. Executive Summary

This report applies a Bayesian attribution model to quantify how strongly observed indicator reuse supports a given candidate cluster (“component”) versus alternative explanations. Using the supplied plug-in likelihood parameters (IP reuse, hash reuse, and a behavioral-overlap indicator) and equal priors across 90 candidate components, the analysis identifies component **1** as overwhelmingly supported by the evidence.

Under the one-vs-rest formulation (component  $i$  responsible vs not responsible baseline), the posterior for component 1 is:  $P(H_1 | D) \approx 1 - 1.30 \times 10^{-912}$  (effectively 1.0).

Under the multi-candidate normalization (one vs many candidates), component 1 has:  $P(H_1 | D) \approx 1 - 8.59 \times 10^{-121}$ .

Important: These probabilities are conditional on the assumptions documented below (indicator independence, parameter choices, and dataset scope). They quantify support under the model; they do not, by themselves, constitute definitive attribution.

## 2. Inputs and Scope

Data sources provided by the requester (local files):

- **ip\_data.xlsx** (sheet “Occurrences”, table=Nodes): IP indicators grouped by component\_id.
- **malicious\_executables\_index.csv**: file/hash indicators grouped by component\_id.

Additional provided files (not directly used in the core computation due to missing component mapping fields): AGP\_overlaps\_master.csv, malicious\_executables\_index.xlsx, top\_malicious\_executables.csv.

Counts derived from the two primary inputs:

- Unique IP indicators:  $n_{ip} = 986$  across 86 components
- Unique executable/hash indicators:  $n_{hash} = 32$  across 5 components
- Components with both IP and hash evidence: 1 (component 1)

## 3. Model Specification

Let each candidate component  $i$  define hypothesis  $H_i$  (“component  $i$  is the responsible actor/cluster”).

Evidence  $D$  is summarized per component as:

- $k_{ip,i}$ : matched IP count (out of  $n_{ip}$ )
- $k_{hash,i}$ : matched hash/executable count (out of  $n_{hash}$ )
- $x_{beh,i} \in \{0,1\}$ : behavioral overlap indicator

Likelihoods (as provided) are modeled as independent conditional on  $H_i$ :

- IP reuse:  $\text{Binomial}(n_{ip}, p)$
- Hash reuse:  $\text{Binomial}(n_{hash}, p)$
- Behavioral overlap:  $\text{Bernoulli}(p)$

Plug-in probabilities used (as supplied in your example):

Under  $H_{\text{res}}$  (responsible):  $p_{\text{ip}}=0.5$ ,  $p_{\text{hash}}=0.4$ ,  $p_{\text{beh}}=0.9$

Under  $H_{\text{nr}}$  (not responsible):  $p_{\text{ip}}=0.05$ ,  $p_{\text{hash}}=0.02$ ,  $p_{\text{beh}}=0.1$

Priors: uniform across components ( $P(H_{\text{res}})=1/90$ ), with  $P(\neg H_{\text{res}})=1-P(H_{\text{res}})$  for one-vs-rest.

## 4. Posterior Computation

Two related posterior views are reported:

### A) One-vs-rest posterior (per component):

$$P(H_{\text{res}}|D) = [L_{\text{res}}(i) \cdot P(H_{\text{res}})] / ([L_{\text{res}}(i) \cdot P(H_{\text{res}})] + [L_{\text{nr}}(i) \cdot P(\neg H_{\text{res}})])$$

where  $L_{\text{res}}(i)$  uses  $(p_{\text{res}}, \text{ip}, p_{\text{res}}, \text{hash}, p_{\text{res}}, \text{beh})$  and  $L_{\text{nr}}(i)$  uses  $(p_{\text{nr}}, \text{ip}, p_{\text{nr}}, \text{hash}, p_{\text{nr}}, \text{beh})$ .

### B) One-vs-many posterior (multi-candidate normalization):

$$P(H_{\text{res}}|D) = [L_{\text{res}}(i) \cdot P(H_{\text{res}})] / \sum_j [L_{\text{res}}(j) \cdot P(H_{\text{res}})]$$

This form compares candidates against each other (not against an explicit baseline  $H_{\text{nr}}$ ).

## 5. Results

| component | idk_ip | k_hash | x_beh | P( $H_{\text{res}} D$ ) one-vs-rest | P( $H_{\text{res}} D$ ) one-vs-many | log10(BF) ( $H_1/H_0$ ) |
|-----------|--------|--------|-------|-------------------------------------|-------------------------------------|-------------------------|
| 1         | 901    | 28     | 1     | 1.000e+00                           | 1.000e+00                           | 913.84                  |
| 421       | 0      | 1      | 0     | 8.730e-284                          | 2.184e-124                          | -281.11                 |
| 418       | 0      | 1      | 0     | 8.730e-284                          | 2.184e-124                          | -281.11                 |
| 403       | 0      | 1      | 0     | 8.730e-284                          | 2.184e-124                          | -281.11                 |
| 203       | 0      | 1      | 0     | 8.730e-284                          | 2.184e-124                          | -281.11                 |
| 326       | 1      | 0      | 0     | 5.078e-284                          | 1.009e-122                          | -281.34                 |
| 333       | 1      | 0      | 0     | 5.078e-284                          | 1.009e-122                          | -281.34                 |
| 332       | 1      | 0      | 0     | 5.078e-284                          | 1.009e-122                          | -281.34                 |
| 331       | 1      | 0      | 0     | 5.078e-284                          | 1.009e-122                          | -281.34                 |
| 330       | 1      | 0      | 0     | 5.078e-284                          | 1.009e-122                          | -281.34                 |

Key finding (component 1):

- $k_{\text{ip},1} = 901$  of 986 IP indicators
- $k_{\text{hash},1} = 28$  of 32 hash/executable indicators
- $x_{\text{beh},1} = 1$  (behavioral overlap present under the report's operational definition)

Computed support:

- One-vs-rest:  $P(H_{\text{res}}|D) \approx 1 - 1.30 \times 10^{-912}$
- One-vs-many:  $P(H_{\text{res}}|D) \approx 1 - 8.59 \times 10^{-121}$
- Bayes factor vs baseline  $H_{\text{nr}}$ :  $\log_{10}(\text{BF}) \approx 913.84$  (decisive under common interpretive scales)

## 6. Interpretation (Forensic Context)

Within the supplied datasets, component 1 contains the overwhelming majority of the IP indicators and hash/executable indicators, and it is the only component exhibiting both categories simultaneously. Given the model's assumption that indicator reuse is much more probable under responsibility than under non-responsibility ( $p_{\text{res}} \gg p_{\text{nr}}$ ), the resulting posterior concentrates on component 1.

This result should be interpreted as: “Under the stated assumptions and parameterization, the observed reuse pattern is vastly more consistent with component 1 than with alternative components or the baseline non-responsibility model.”

## 7. Assumptions and Limitations

- **Indicator independence:** IP reuse, hash reuse, and behavior are treated as conditionally independent; real-world dependencies can inflate confidence.
- **Parameter sensitivity:** p-values ( $p_{IP}$ ,  $p_{hash}$ ) are analyst-selected plug-ins; different values can materially change the posterior.
- **Dataset scope:** Results reflect only the provided files; missing telemetry or biased sampling can skew counts.
- **Operational behavior definition:**  $x_{beh}$  is derived as “both IP and hash evidence present.” If behavioral overlap is defined differently, results may differ.
- **Attribution caution:** High posterior under this model is not proof of authorship or sponsorship; it is a quantitative consistency measure.

## Appendix A — Reproducibility Notes

Deterministic counts:

$n_{ip}$  = unique(ip) from ip\_data.xlsx / Occurrences where table=Nodes

$n_{hash}$  = unique(entity\_id) from malicious\_executables\_index.csv

Per component:

$k_{ip,i}$  = unique(ip) by component\_id

$k_{hash,i}$  = unique(entity\_id) by component\_id

$x_{beh,i} = 1$  iff  $k_{ip,i} > 0$  and  $k_{hash,i} > 0$  else 0